

Transformer-Based News Summarization Using BART

MSCS2201-1: Artificial Intelligence

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<https://github.com/dkim4741/ai-mini-research-problem>

<https://youtu.be/5gE61xNXGu4?si=pl3nGFIS2f2PeJpt>

tl;dr

- **Motivation:** Long news articles require significant reading time. Automatic summarization can help users quickly understand key information while reducing information overload.
- **Main idea:** Use a pre-trained BART Transformer model from Hugging Face to automatically summarize long news articles and analyze summarization performance.
- **Result:** The system successfully generated concise summaries while preserving the main ideas of the original articles. Test articles were compressed to approximately 10–35% of their original length.

Literature Review (Related Methods)

- Transformers (Vaswani et al., 2017):
 - Introduced the Self-Attention mechanism, replacing traditional RNNs/CNNs for sequence modeling.
 - Allows the model to capture long-range dependencies and contextual relations between words in parallel.
- BART (Lewis et al., 2019):
 - A Sequence-to-Sequence (Seq2Seq) model that combines a bidirectional encoder (like BERT) and an autoregressive decoder (like GPT).
 - Pre-trained as a Denoising Autoencoder, making it exceptionally strong for text generation and summarization tasks.

Approach & Implementation

- Model: Pretrained BART (e.g., facebook/bart-large-cnn) from Hugging Face.
- Steps:
 - User pastes a news article into the Gradio interface.
 - The article is processed by the pre-trained BART model.
 - BART generates an abstractive summary.
 - The system calculates:
 - Original Word Count
 - Summary Word Count
 - Compression Ratio
 - Results are displayed through the web interface.
- Code:
 - <https://github.com/dkim4741/ai-mini-research-problem>

System Workflow

News Article

- Gradio Web Interface
- BART Transformer Model
- Generated Summary
- Word Count & Compression Ratio Analysis

Experimental Results

Test Articles	Original Words	Summary Words	Compression	Result
Normal	472	88	18.64%	Successful
Long	696	77	11.06%	Successful
Noisy	300	100	33.33%	Robust
Mixed Format	289	90	31.14%	Acceptable
Very Short	52	75	144.23%	Failed

Normal: Accurately captured key names and events with high fluency.

Long: Maximized compression efficiency without losing core timelines.

Noisy: Demonstrated word-level robustness by auto-correcting typos.

Mixed Format: Successfully converted fragmented bullet points into prose.

Very Short: Forced text expansion (144.23%) due to strict token constraints.

BART tokens: 100-200 (80-170 words)

Note: Due to the input token limitation of the BART model, all test articles were limited to approximately 700 words or fewer.

Demo & Conclusion

- Demo video:
 - <https://youtu.be/5gE61xNXGu4?si=pl3nGFIS2f2PeJpt>
- What the demo shows:
 - Various versions of articles (normal, long, noisy, mixed format, very short)
 - Real-time text summarization process via Gradio web interface
 - Statistical analysis reporting: Original vs. Summary word counts and calculated compression ratios (%)
- Takeaways:
 - Gradio web UI provides an intuitive, hands-on framework to evaluate natural language generation models in real time.
 - Pre-trained BART model demonstrates strong contextual compression capabilities without the need for high computing training costs.
- Limitations:
 - Strict minimum token constraints (min_length) can force severe over-generation and hallucination when processing very short input texts.

References

1. Theoretical Background (Literature Review)

- Vaswani, A., et al. (2017). "Attention Is All You Need." Advances in Neural Information Processing Systems (NeurIPS), 30. (Introduced the Transformer architecture and Self-Attention mechanism)
- Lewis, M., et al. (2019). "BART: Denoising Sequence-to-Sequence Pre-training for Natural Language Generation, Translation, and Comprehension." arXiv preprint arXiv:1910.13461. (The core architecture used for this summarization system)

2. Experimental Data (Articles used in Case Studies)

- [Case 1: Normal] Times of India (2024). "NASA announces crew for Artemis III mission, paving way to lunar south pole mission."
- [Case 2: Long] NASA JPL / The Robot Report (2021). "How NASA's Perseverance Rover made oxygen on Mars."
- [Case 3: Noisy] Robustness Test Sample. Derived from Case 2 (NASA MOXIE News) and modified with synthetic noise (typos, slang, and informal structure) for model evaluation.
- [Case 4: Mixed Format] World Economic Forum (2024). "Global Cybersecurity Outlook 2024 Report."
- [Case 5: Very Short] Live Science (2026). "China's 'artificial sun' reactor shatters major fusion limit."